

Quick transformations

Always think about a scale that makes the model easier to interpret!

Shade level:

-1 : Low

0: Medium

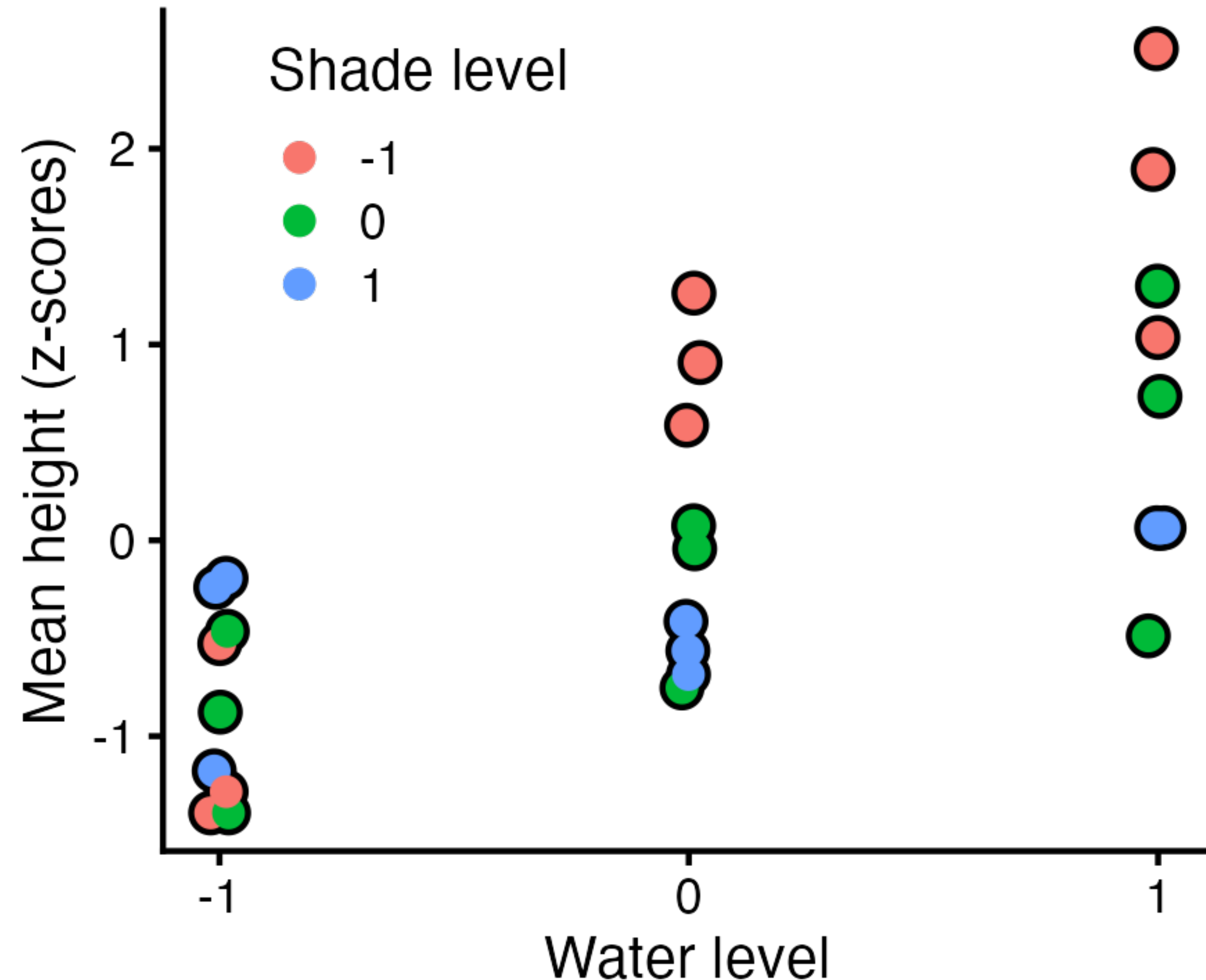
1: High

Water level:

-1 : Low

0: Medium

1: High



Model with interaction

$Height_i \sim Normal(\mu_i, \sigma)$

$$\mu_i = a + b \text{ water}_i + c \text{ shade} + d \text{ water}_i \times \text{shade}_i$$

```
data("tulips")
df = tulips
df$water  = scale(df$water, scale = FALSE)
df$shade  = scale(df$shade, scale = FALSE)
df$blooms = scale(df$blooms)

rt_fit = ulam(alist(
  blooms ~ normal(mu, sigma),
  mu <- a + b*water + c*shade + d*water*shade),
  data = df)
```

